MIT SCHOOL OF COMPUTING
Rajbaug, Loni-Kalbhor, Pune

MIT-ADT UNIVERSITY
PUNE, INDIA
A Leap Towards World Class Education

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Subject: Roll No.: 28 Exp. No.: 1

Name of the Experiment:

Performed on:

Submitted on:

Marks	Teacher's Signature with date
10	

Q.1 List the key features and advantages of python. Also mention any two python IDES or editors used for program development.

→ key features of python

1. Python is a Simple and easy to learn programming language.
2. It is an interpreted language, so code is executed line by line
3. Python is a high-level language, which makes programming easier.
4. It supports object-oriented programming concepts.
5. Python is platform independant.
6. It is open-source language an free to use.
7. Python has a large standard library.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

* Advantages of python.

- Requires less code compared to other language
- Easy debugging and maintenance.
- used in web development, data science, AI, and automation.
- Suitable for beginners as well as Professionals.

* Any two python IDES/Editors

1. Pycharm

2. Visual Studio code (VS)

1.2 Explain the history and evolution of python. Describe why python has become popular for beginners and Industry applicaton

→ History of python

Python was developed by Guido von Rossum in 1989 and released in 1991. The name "Python" was inspired by the comedy show monty python's flying circus.

* Evolution of python

- Python 3.x: Improved version with better performance, security, and unicode Support.

- Python 2.x: older version

* why python is popular

- Easy Syntax Similar to english
- widely used in industries
- large community support.
- used by companies like google, Netflix and Instagram.
- Supports multiple domains such as AI, ML, web development, and Cyber security.

Q.3 write a python program to accept two numbers and perform arithmetic operations.

→ program:

```
a=Float(input("enter first number:"))
b=Float(input("enter Second number:"))
```

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```
Print("Addition=",a+b)
Print("substraction=",a-b)
Print("Multiplication=",a*b)
Print("Divison=",a/b)
```

Explanation: The program accepts two number values from the user and Performs addition, Subtraction, Multiplication and division.

Q.4 Analyze the python code and identify data types. Explain the output

→

```
a=10
b=5.5
c="python"
d= True
```

Data types
a ⇒ Integer
b ⇒ Float
c ⇒ String
d ⇒ Boolean

Explanation:- Python automatically assigns data types based on the value stored in the variable.

5 Evaluate relational, logical, and bitwise operators with examples where each type of operator is most appropriate.

→

Relational operators.

used to Compare values. Code :-

```
a=10
b=5
Print(a>b)
```

used in decision-making Statements.

logical operators :

used to combine multiple conditions
Code:-

```
X=True
Y=False
Print(X and Y)
```

used in Conditional logic.

Bitwise operators:

used in bit-level operations Code

```
a=5
b=3
Print(a&b)
```

used in low-level programming

Q.6 Design and implement a menu-driven python program that.

> Accepts user input
> Performs arithmetic, relational, lo and assignment operation.
> Displays results clearly
The program should handle int float and Boolean data types.

→ Code:-

```
a = Int(input("Enter first number:"))
b = Int (input("Enter second number

Print("1. Arithmetic operation")
Print("2. Relational operation")
Print("3. logical operation")
Print("4. Assignment operation")

Choice = int(input("Enter your choice:

if Choice = 1;
    Print ("Addition=", a+b)
    elif Choice == 2;
    Print ("a>b =", a>b)
    elif Choice == 3;
    Print((a>0) and (b>0))
```

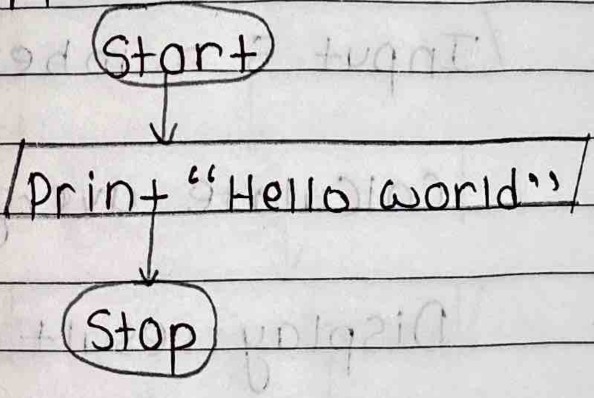
```
elif Choice = 4;
        a+=b
Print("result after assignment:",a)
 else:
Print("Involid Choice")
```

LAB ASSIGNMENT

Q.7 write a program in python to display "Hello world".

→ Algorithm:
1. Start
2. Print "Hello world"
3. Stop

Flowchart



: python Code: Print("Hello world")

output: Hello world

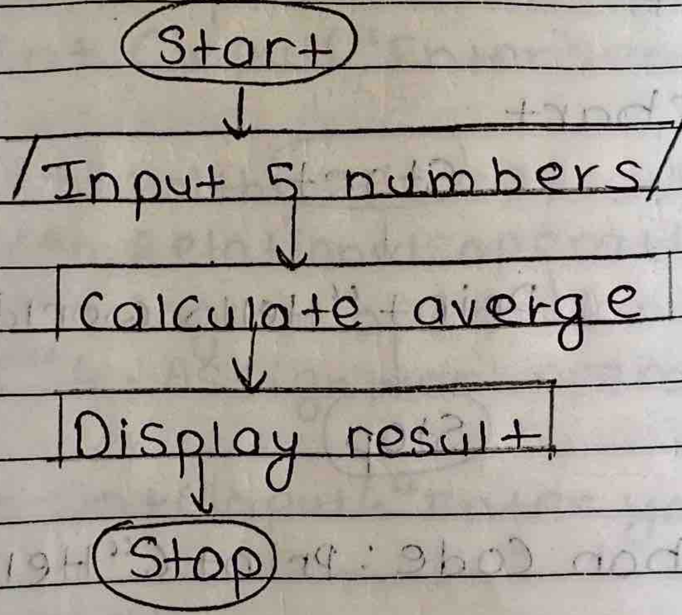
Q 8 write a program to perform ave
of five numbers using input()
Ction and type conversion co

→

Algorithm:

1. Start
2. Input five numbers
3. Convert to float
4. Calculate average
5. Display result.
6. Stop.

Flow chart



Python Code:

```
a = Float(Input("Enter number1: "))
b = Float(Input("Enter number2: ")
c = Float(Input("Enter number 3: ")
```

```
d=Float(input("Enter number 4:"))
e=Float(input("Enter number 5:"))

avg=(a+b+c+d+e)/5
Print("average e=",avg)
```

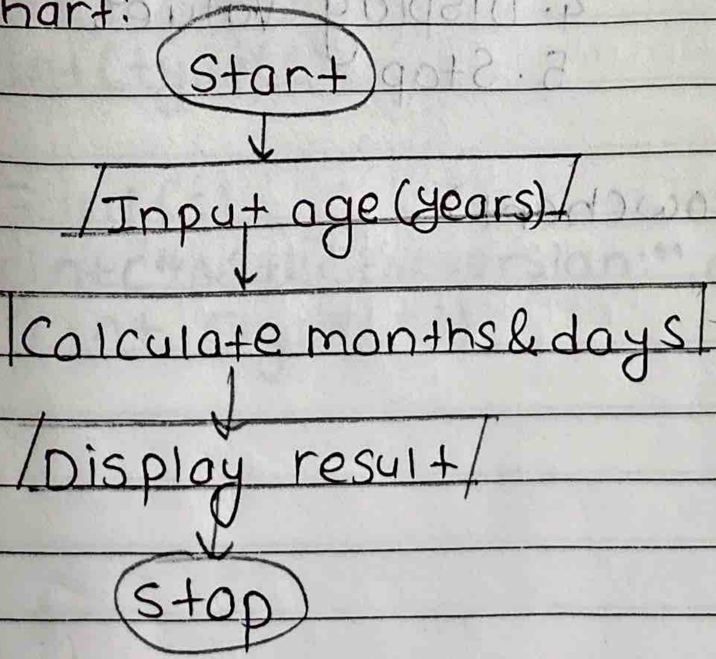
write a program that prompts the user to enter their age and prints out their age in years, months, and days.

2

Algorithm:

1. Start
2. Input age in years
3. Convert to months and days
4. Display result
5. Stop

Flowchart:



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Python Code:

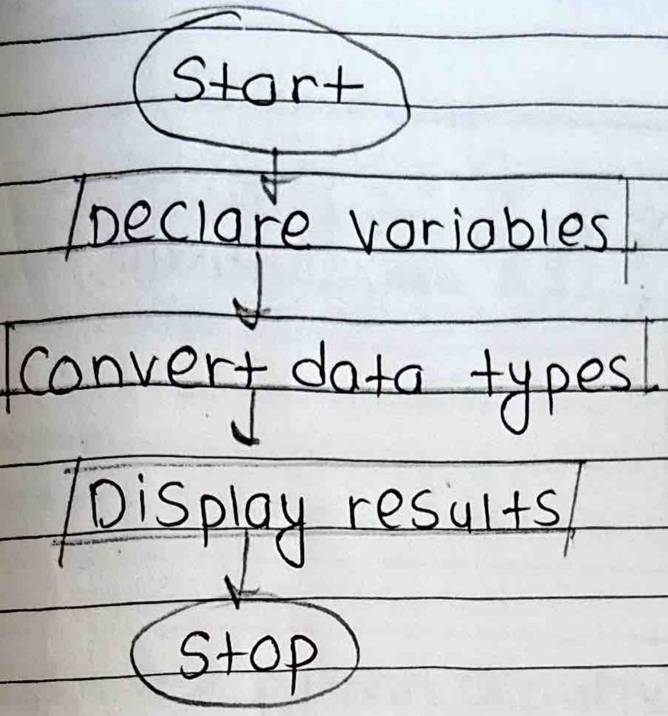
```
age= int(input("Enter your age
                          years:"))

  Print("Age in years:" age)
  Print("Age in months:" months)
  Print("Age in Days:" days)
```

Q.10 write a program to demonstrate the concept of data typs & conversi

→

Algorithm:
1. Start
2. Declare variables of different data types
3. Convert data types
4. Display values
5. Stop

Flowchart:



Python Code

```
a=10
b=5.5
c="20"

Print(type(a))
Print (type(b))
Print(type(c))

c= int(c)
Print("After conversion:",c)
Print (type(c))
```